

Ethical models: How rule-based and Bayesian approaches influence the development of Machine Learning and Artificial Intelligence

MADS – Ethics assignment 1

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Introduction

Throughout the years, The use of Machine Learning (ML) and Artificial Intelligence (AI) is ever evolving. Thus so must the methods used behind building the algorithms of aforementioned intelligent software. Within its development, ethics has played a major role in the course of direction. Methods in which ethics has been incorporated can be divided in two directions, rule-based and Bayesian.

Rule-based ethics in models

When speaking of rule-based models, systems that make decisions based on a predefined set of rules . These rules are often applied in an *if-then* manner, wherein *if* stands for specific conditions and *then* triggers the corresponding result (Milvus, 2026).

Multiple sources state that over the course of time, rule-based models have become too rigid and therefore incapable of handling modern Question Answering (QA) regarding moral uncertainty. It turns out that human language is too complex to capture in a finite set of if-then rules. Even though rules are safe, they are brittle. Furthermore, if ethical rules are hardcoded in standard models it can still result in harmful answering.

Assuming within this statement that there is a difference in ethical movement, the asked question being harmful or biased and utilizing a standard model (Abel, MacGlashan, & Littman, 2016), (Sipio, 2025).

Bayesian ethics in models

When utilizing Bayesian models, instead of hard-coded rules, the algorithm updates its beliefs based on new evidence, thus making decisions based on probability (The Statsig Team, 2025).

Bayesian models can be used to manage moral uncertainty by prioritizing information gathering to navigate ethical ambiguity. (Abel, MacGlashan, & Littman, 2016) and (Sipio, 2025) state that in order for an agent to be ethical, it should maintain a probability distribution over which ethical decisions are made. By incorporating epistemic uncertainty, a trustworthy AI that can recognize its own limitations and abstain from answering, thus taking accountability can be formed. Herein, epistemic uncertainty is defined as systemic uncertainty due to limitations in data, measurement precision, lack of knowledge or a multitude of aforementioned factors (FORRT, 2026).

Hybrid models, an interim method

Both methods aid in incorporating ethics in ML and AI. They differ in the sense that rule-based is set black-and-white reasoning and Bayesian is more of an evolving gray area. As well as used separately, rule-based and Bayesian models can be combined to create a Hybrid model.

(Pelluru, 2025) advocates using Bayesian logic to provide justifications for decisions, acting as a manner of explaining ethical reasoning behind traditional rule-based systems. Conversely, within a hybrid system, rule-based constraints can act as a safety measure around probabilistic learning models.

Regarding data ethics

From my standpoint, models should be able to continuously update their decision making, preventing model drifting and keeping up with the times. This will prevent erroneous QA and decision making by the model becoming outdated. Furthermore, it should be clear that ethics strongly rely on nature, nurture and culture and can therefore differ based on stakeholder.

For that reason, the most suitable model according to my opinion is the use of hybrid models in which ethics are incorporated in both rule-based and Bayesian algorithms, which can evolve, keep their counterparts in check, but can also refrain from generating a response if there is uncertainty. Uncertainty is a form of honesty which I am a fervent advocate of.

Bibliografie

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